

Classification of very stable Higgs bundles

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Very stable Higgs bundles, equivariant multiplicity and mirror symmetry

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Abstract We define and study the existence of very stable Higgs bundles on Riemann surfaces, how it implies a precise formula for the multiplicity of the very stable components of the global nilpotent cone and its relationship to mirror symmetry. The main ingredients are the Bialynicki-Birula theory of $\mathbb{C}^*$ -actions on semiprojective varieties, $\mathbb{C}^*$ characters of indices of $\mathbb{C}^*$ -equivariant coherent sheaves, Hecke transformation for Higgs bundles, relative Fourier–Mukai transform along the Hitchin fibration, hyperholomorphic structures on universal bundles and cominuscule Higgs bundles.

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Main goal

Study and classify **very stable Higgs bundles**.

- Moduli space of polystable G -Higgs bundles $\mathcal{M}(G)$.
- Complex quasiprojective variety with $\mathbb{C}^\times$ -action.
- **Semiprojective:** Given a point in $\mathcal{M}(G)$, the scaling limit at 0 always exists, and is a fixed point $\mathcal{E}$.

$$W_{\mathcal{E}}^+ = \{\text{Points whose limit at 0 is } \mathcal{E}\} \subseteq \mathcal{M}(G)$$

is the **upward flow** of the fixed point $\mathcal{E}$.

Definition (Hausel–Hitchin, 2022)

A smooth fixed point $\mathcal{E}$ is **very stable** if and only if $W_{\mathcal{E}}^+$ is **closed**. Otherwise, **wobbly**.

- Why? **Topology, geometry and duality/physics** of $\mathcal{M}(G)$.
- **Topology/geometry of $\mathcal{M}(G)$. Hitchin map**

$$h_G : \mathcal{M}(G) \rightarrow \mathcal{A}(G),$$

to vector space, proper, surjective, $\mathbb{C}^\times$ -equivariant.

- $\mathcal{M}(G)$ retracts to $h_G^{-1}(0)$: **global nilpotent cone**.
- $\mathbb{C}^\times$ acts on projective (singular) $h_G^{-1}(0)$, **very stable** Higgs bundles $\leftrightarrow$ **trivial upward flows**.
- Consider $h_G|_{W_\mathcal{E}^+} : W_\mathcal{E}^+ \rightarrow \mathcal{A}(G)$.
- **Very stable** Higgs bundles $\leftrightarrow$ **finite cover** (finite, flat, surjective, generically unramified).

- **Physics of $\mathcal{M}(G)$.** Mirror symmetry: dual variety $\mathcal{M}(G^\vee)$.
- Dual $\rightsquigarrow$ several statements explained by physics (e.g. Kapustin–Witten 2007), also geometric Langlands program (e.g. Beilinson–Drinfeld 1995, Donagi–Pantev 2012).
- Prediction: correspondence between **BAA-branes** on $\mathcal{M}(G)$ and **BBB-branes** on $\mathcal{M}(G^\vee)$.
- Fact: $\mathcal{E}$ smooth fixed point $\rightsquigarrow W_{\mathcal{E}}^+$ complex Lagrangian $\rightsquigarrow$ support of BAA-brane.
- **Very stable** Higgs bundles $\leftrightarrow \mathbb{C}^\times$ -invariant, **closed, affine** irreducible complex Lagrangian subvarieties.

Higgs bundles

- G connected semisimple algebraic group over $\mathbb{C}$, Lie algebra $\mathfrak{g}$.
- C smooth projective curve over $\mathbb{C}$, genus $g \geq 2$, canonical line bundle K_C .

Definition

A **G -Higgs bundle** is a pair (E, φ) :

- Principal G -bundle $E \rightarrow C$
 - $\varphi \in H^0(C, E(\mathfrak{g}) \otimes K_C)$
-
- $\mathcal{M}(G)$ parametrises isomorphism classes of **polystable** G -Higgs bundles.
 - **Example.** $G = \mathrm{PGL}_n(\mathbb{C}) \rightarrow$ pair of **rank n vector bundle** E (up to line bundle tensoring) and $\varphi : E \rightarrow E \otimes K_C$ traceless.

- $\lambda \in \mathbb{C}^\times$ acts: $(E, \varphi) \mapsto (E, \lambda\varphi)$.
- **Fixed points.** E semistable $\rightarrow (E, 0) \in \mathcal{M}(G)^{\mathbb{C}^\times}$.
- Others: e.g. $G = \mathrm{PGL}_n(\mathbb{C})$, $E = E_0 \oplus \cdots \oplus E_{m-1}$,

$$\varphi = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \varphi_1 & 0 & \cdots & 0 & 0 \\ 0 & \varphi_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \varphi_{m-1} & 0 \end{pmatrix}$$

- In general, theory of **Vinberg** $\mathbb{C}^\times$ -pairs, representations of Levi subgroups $G_0 \subseteq G$.

Fixed points and gradings

$\mathbb{Z}$ -gradings:

$$\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j$$

with $[\mathfrak{g}_j, \mathfrak{g}_k] \subseteq \mathfrak{g}_{j+k}$.

$[\mathfrak{g}_0, \mathfrak{g}_0] \subseteq \mathfrak{g}_0 \rightsquigarrow G_0 \subseteq G$ connected subgroup.

$[\mathfrak{g}_0, \mathfrak{g}_j] \subseteq \mathfrak{g}_j \rightsquigarrow$ **representation** $G_0 \rightarrow \mathrm{GL}(\mathfrak{g}_j)$.

Prop. (Simpson 1988, Biquard–Collier–García-Prada–Toledo 2023)

Fixed points $(E, \varphi) \in \mathcal{M}(G)^{\mathbb{C}^\times}$ characterised by:

- E reduces to E_{G_0} a G_0 -bundle.
- $\varphi \in H^0(E_{G_0}(\mathfrak{g}_j) \otimes K_C)$ for $j \neq 0$

Previous results

- $G = \mathrm{GL}_n(\mathbb{C})$. Fixed points $(E, 0)$, E semistable. (Laumon, 1988): very stable vector bundle, open dense condition.
- $G = \mathrm{GL}_n(\mathbb{C})$, $E = L_0 \oplus \cdots \oplus L_{n-1}$, $\mathrm{rank} L_i = 1$:

Theorem (Hausel–Hitchin 2022)

(E, φ) very stable if and only if $\varphi_{n-1} \circ \cdots \circ \varphi_1 : L_0 \rightarrow L_{n-1} \otimes K_C^{n-1}$ is a reduced divisor (no multiple zeroes).

- $G = \mathrm{GL}_n(\mathbb{C})$, $E = E_0 \oplus E_1$: (Peón-Nieto, 2024) identifies components without very stable Higgs bundles.

Generically regular Higgs field

- φ generically regular $\rightsquigarrow$ rep. $(G_0, \mathfrak{g}_1)$ where $\mathfrak{g}_1$ has regular elements.
- Forces $G_0 = T$ maximal torus, $\mathfrak{g}_1 = \bigoplus_{j=1}^r \mathfrak{g}_{\beta_j}$, with $\{\beta_j\}_{j=1}^r \subseteq \mathfrak{t}^*$ simple roots for Borel $B \supseteq T$.
- Fixed points of **Borel type**.
- Projection

$$E_T(\mathfrak{g}_1) \otimes K_C \rightarrow E_T(\mathfrak{g}_{\beta_j}) \otimes K_C$$

gives sections φ_j .

- (Recall $E = L_0 \oplus \cdots \oplus L_{n-1}$, divisors $\varphi_j: L_{j-1} \rightarrow L_j \otimes K_C$).

Classification theorem for Borel type

- **Multiplicity divisor** $\mu(E, \varphi)$ valued in coweight lattice $P^\vee = \{\mu : \mathbb{C}^\times \rightarrow T_{\text{ad}}\} \subseteq \mathfrak{t}$.
- Defined by

$$\beta_j(\mu(E, \varphi)) := \text{div}(\varphi_j).$$

- **Reduced** if every coefficient **minuscule** (minimal under Bruhat/dominance order on $P_+^\vee$).

Theorem (G. 2025)

A smooth fixed point $(E, \varphi) \in \mathcal{M}(G)^{s\mathbb{C}^\times}$ of Borel type is very stable if and only if $\mu(E, \varphi)$ is reduced.

- Conjecture 8.2 in (Hausel–Hitchin, 2022).

- If $\mu(E, \varphi)_c$ is not minuscule, use **Hecke transformations** to produce a $\mathbb{C}^\times$ -orbit in $W_{(E, \varphi)}^+ \cap h_G^{-1}(0)$.

Definition

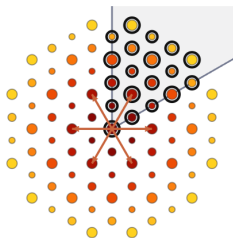
A **Hecke transformation** of (E, φ) at $c \in C$ is another (E', φ') with isomorphism over $C_0 := C \setminus \{c\}$:

$$(E', \varphi')|_{C_0} \xrightarrow{\sim} (E, \varphi)|_{C_0}$$

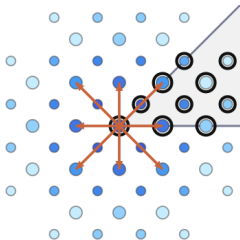
- Without Higgs field, parametrised by **affine Grassmannian** $\mathrm{Gr}_G = G((z))/G[[z]]$.
- Locally, Higgs field $\varphi_1 \in \mathfrak{g}[[z]]$. Hecke transformations given by affine Springer fibre $\mathrm{Gr}_G^{\varphi_1}$.
- **Main idea:** $\mathbb{C}^\times$ -action on Hecke transformation spaces $(\mathrm{Gr}_G$ and $\mathrm{Gr}_G^{\varphi_1})$.

Strategy II: Fixed points in Gr_G

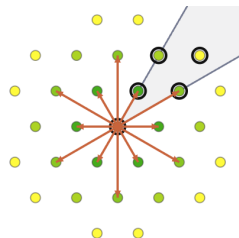
Fixed points: $\text{Gr}_T = X_*(T) = \{\mu : \mathbb{C}^\times \rightarrow T\} \hookrightarrow \text{Gr}_G$.



$G = \text{PGL}_3(\mathbb{C})$



$G = \text{SO}_5(\mathbb{C})$



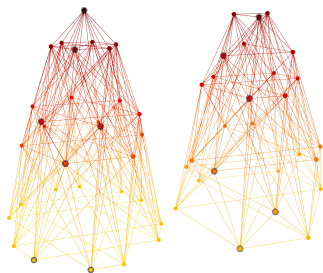
$G = G_2$

Hecke transformation by $\mu \in X_*(T)$ gives another fixed point (E', φ') of Borel type with

$$\mu(E', \varphi') = \mu(E, \varphi) - \mu \cdot c.$$

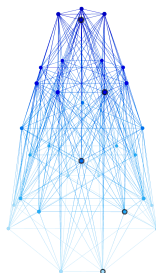
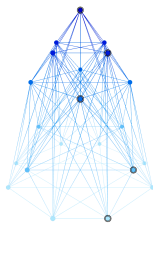
Strategy III: $\mathbb{C}^\times$ -orbits in Gr_G

Generate $\mathbb{C}^\times$ -orbits in $\mathcal{M}(G)$ by using $\mathbb{C}^\times$ -orbits in Gr_G . Easiest:
affine root subgroups shifted by $X_*(T)$.

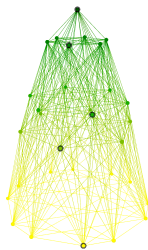


x2

$$G = \mathrm{PGL}_3(\mathbb{C})$$



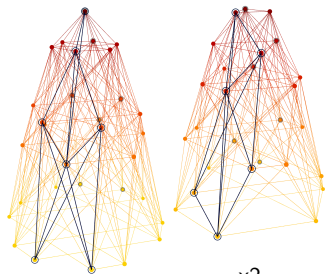
$$G = \mathrm{SO}_5(\mathbb{C})$$



$$G = G_2$$

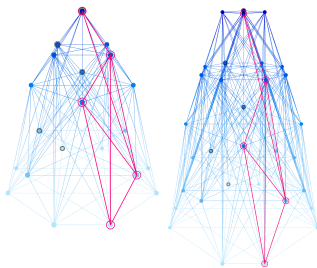
Strategy IV: $\mathbb{C}^\times$ -orbits in Springer fibre

Also need to consider stability, and Springer fibres.

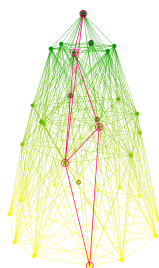


x2

$$G = \mathrm{PGL}_3(\mathbb{C})$$



$$G = \mathrm{SO}_5(\mathbb{C})$$



$$G = G_2$$

- If every $\mu(E, \varphi)_c$ is minuscule, any $\mathbb{C}^\times$ -orbit in $W_{(E, \varphi)}^+ \cap h_G^{-1}(0)$ **comes from Hecke transformations** of (E', φ') with $\mu(E', \varphi') = \mu(E, \varphi) - \mu(E, \varphi)_c \cdot c$.
- Must come from

$$G[[z]]z^{\mu(E, \varphi)_c} \subseteq \mathrm{Gr}_G,$$

intersects the Springer fibre at a single point.

- Thus (E, φ) is very stable.

Parabolic case and fixed points

- Strategy extends to **strongly parabolic Higgs bundles**.
- **Punctures** $D = c_1 + \cdots + c_s$ in C
- **Parabolic structures** $Q_i \in E|_{c_i}/B$.
- **Higgs field** with **simple poles** at D and nilpotent wrt Q_i .
- **Moduli space** $\mathcal{M}_{sp}(G, \alpha)$ with similar properties. (e.g. Simpson 1990, Konno 1993, Yokogawa 1993, Markman 1994, Bottacin 1995, Biquard–García-Prada–Mundet i Riera 2020)

Proposition

Fixed points $(E, \varphi, Q) \in \mathcal{M}_{sp}(G, \alpha)^{\mathbb{C}^\times}$ characterised by:

- E reduces to E_{G_0} a G_0 -bundle.
- $\varphi \in H^0((E_{G_0}(\mathfrak{g}_j) \cap \text{SPE}(\mathfrak{g})) \otimes K_C(D))$ for $j \neq 0$.
- $Q_i \in \bigsqcup_{w \in W_{G_0} \setminus W} (E_{G_0}|_{c_i} \cdot w \cdot B)/B \subseteq E|_{c_i}/B$.

Parabolic fixed points of Borel type

- Borel type: $(T, \mathfrak{g}_1 = \bigoplus_{j=1}^r \mathfrak{g}_{\beta_j})$ forces

$$Q_i \in \bigsqcup_{w \in W} (E_T|_{c_i} \cdot w \cdot B)/B \simeq W.$$

- **Twisted multiplicity divisor** $w(E, \varphi, Q)$ valued in **extended affine Weyl group**:

$$\widetilde{W} = P^\vee \rtimes W.$$

- **Reduced**: every coefficient minimal under Bruhat order.

Theorem (G. 2026)

A smooth fixed point $(E, \varphi, Q) \in \mathcal{M}_{sp}(G, \alpha)^{s\mathbb{C}^\times}$ of Borel type is very stable if and only if $w(E, \varphi, Q)$ is reduced.

- Hecke transformations at $c_i \in D$ given by (affine Springer fibres in) **affine flag variety** $\mathrm{Fl}_G := G((z))/\mathcal{I}$ where $\mathcal{I} \subseteq G[[z]]$ is the **standard Iwahori subgroup**.
- Similar strategy using Fl_G instead of Gr_G .
- **Fixed points** are $\tilde{W} \hookrightarrow \mathrm{Fl}_G$.
- **Two kinds** of $\mathbb{C}^\times$ -orbits using $\mathrm{Fl}_G \rightarrow \mathrm{Gr}_G$ with fibre G/B .

Non-regular case (non-parabolic)

- More general $(G_0, \mathfrak{g}_1)$. **Toledo invariant**

$$-(2g-2) \operatorname{rank}_T(G_0, \mathfrak{g}_1) \leq \tau(E, \varphi) \leq 0$$

defined in (Biquard–Collier–García-Prada–Toledo, 2023).

- Useful to exclude very stable Higgs bundles. For $\mathfrak{g} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1$, where $\tau(E, \varphi) \sim \deg E_{G_0}(\mathfrak{g}_1)$:

Theorem (García-Prada–G. 2026)

Suppose that

$$\tau(E, \varphi) < 2B^*(\beta, \beta)(g-1) \left(\frac{-\operatorname{rank}_T(\varphi)}{\hat{B}^*(\hat{\beta}, \hat{\beta})} + \dim \mathfrak{g}_1 - \dim \hat{\mathfrak{g}}_1 \right).$$

Then (E, φ) is not very stable.

Non-regular case (non-parabolic) II

- In the previous theorem, nonreductive G_0^e for $e \in \mathfrak{g}_1$ generic is required.
- Otherwise we can exploit the Cayley correspondence (e.g. Biquard–García-Prada–Rubio 2017).

Theorem (García-Prada–G. 2026)

Suppose that $\mathfrak{g} \neq \mathfrak{sl}_2(\mathbb{C}), \mathfrak{sp}_{2n}(\mathbb{C})$, that G_0^e as above is reductive and

$$\tau(E, \varphi) = -(2g - 2) \operatorname{rank}_T(G_0, \mathfrak{g}_1).$$

Then (E, φ) is not very stable.

Additional questions I

- (E, φ) of Borel type with reduced $\mu(E, \varphi)$. Finite cover $h|_{W_{(E, \varphi)}^+}$?
- **Order:** $\text{rank}(h_G)_*(\mathcal{O}_{W_{(E, \varphi)}^+})$, can be computed from its equivariant Euler characteristic $m_{(E, \varphi)}(t) \in \mathbb{Z}((t))$.

Proposition (G., 2025)

We have

$$m_{(E, \varphi)}(t) = \prod_{c \in C} \mathcal{D}_{\mu(E, \varphi)_c}(t)$$

where $\mathcal{D}_\mu(t)$ is the **Dynkin polynomial** for μ .

- In particular, the order is $m_{E, \varphi}(1) = \prod_{c \in C} \dim(V^{\mu(E, \varphi)_c})$.
- (Hausel; G. 2026) If $\mu(E, \varphi) = \mu \cdot c$, pulled from $\mathfrak{t}/W_\mu \rightarrow \mathfrak{t}/W$.

Additional questions II

- Dual BBB-brane to very stable $W_{(E,\varphi)}^+$ of Borel type?
- Generically Fourier–Mukai via dual Hitchin systems (Donagi–Pantev, 2012)

Conjecture

The dual BBB-brane is given by

$$\bigotimes_{c \notin D} \rho^{\mu(E,\varphi)_c}(\mathbb{E}^\vee|_{\mathcal{M}(G^\vee) \times \{c\}}) \otimes \bigotimes_{c_i \in D} \mu(E,\varphi)_{c_i}^{-1}((\mathbb{E}^\vee|_{\mathcal{M}(G^\vee) \times \{c_i\}})^{B^\vee}).$$

- Rank agrees with order from before.
- (Hausel–Hitchin 2022) check Fourier–Mukai transform for $G = \mathrm{GL}_n(\mathbb{C})$, $D = 0$.
- (G. 2026) checks it for $G = \mathrm{PGL}_n(\mathbb{C})$ in parabolic setting, with $\mathrm{supp}(w(E, \varphi, Q)) \subseteq D$.

Happy birthday, Nigel!

